Biochemical Engineering Journal 167 (2021) 107873

Contents lists available at ScienceDirect

Biochemical Engineering Journal

journal homepage: www.elsevier.com/locate/bej

Regular article

Performance evaluation and microbial community analysis of microaerobic pretreatment on thermophilic dry anaerobic digestion

Feng Zhen [a,b,c,d], Xinjian Luo [b,c,d], Tao Xing [b,c,d], Yongming Sun [b,c,d], Xiaoying Kong [b,c,d,**], Wenzhe Li [a,e,*]

[a] College of Engineering, Northeast Agriculture University, Harbin, 150030, China
[b] Guangzhou Institute of Energy Conversion, Chinese Academy of Sciences, Guangzhou, 510640, China
[c] CAS Key Laboratory of Renewable Energy, Guangzhou, 510640, China
[d] Guangdong Provincial Key Laboratory of New and Renewable Energy Research and Development, Guangzhou, 510640, China
[e] Key Laboratory of Agricultural Renewable Resources Utilization Technology and Equipment in Cold Areas of Heilongjiang Province, Harbin, 150030, China

ARTICLE INFO

Keywords:
Microaerobic pretreatment
Dry anaerobic digestion
Microbial community
Corn stalk
Methane production

ABSTRACT

In this study, microaerobic pretreatment (MP) was used to improve the system stability and biogas production performance of the thermophilic dry anaerobic digestion (AD). The results showed that proper MP made the AD have better methane-producing performance and the maximum cumulative methane production was 108.31 mL/g VS with an oxygen load of 5 mL/g VS, which was 13.20 % higher than that of control group (G1, 95.68 mL/g VS). Specific microaerobic conditions promoted the generation of volatile fatty acid (VFA) and reduced propionic acid content in the digestive system. In addition, the volatile solid (VS) removal rate of the digestive substrates was improved in the presence of limited oxygen, which can reduce environmental risks. Microbial community diversity indicated that MP increased the relative abundance of *Firmicutes* to promote substrate hydrolysis and accelerated the growth of hydrogenotrophic methanogens to improve AD performance.

1. Introduction

As an environmentally friendly and efficient method, AD technology has been widely used in the disposal of organic waste such as food waste, sludge, crop stalk and animal manure, which can convert organic matter into methane energy via microorganisms [1–3]. At present, three major AD technologies have been studied: wet (TS $\leq$ 10 %), hemi-dry (10 % $<$TS $<$ 15 %) and dry (TS $\geq$ 15 %) AD, also called high-solid AD [4]. Generally, hemi-dry or dry AD is more attractive with less water, smaller digester and higher volume biogas production. Whereas, it is less efficient than wet AD for mass transfer, heat transfer, and stirring in anaerobic digester [5]. Therefore, low microbial access to substrates and potential accumulation of toxicity and inhibitors (e.g. volatile fatty acids and ammonia) may occur in dry AD, which is detrimental to the digestive system [6].

Previous studies have demonstrated that the hydrolysis of the digestive substrate could be enhanced by the introduction of limited oxygen [7]. Tsapekos, et al. [8] showed that the maximum production of biogas was obtained by injecting 5 mL O_2/g VS for 3 days, which was 7.2 % higher than that of the untreated wheat straw. Lim, et al. [9] also reported that there was an increase in the hydrolysis rate of AD due to greater diversity of bacteria and higher abundance of *Firmicutes* in microaerobic conditions. The potential mechanism of enhanced AD may be that the addition of microoxygen inhibit the formation of toxic metabolites (such as lactic acid and ethanol) and promote the synthesis of substances required for the stability of anaerobic cell membranes [10]. The rapid growth of facultative bacteria can produce more cellulolytic and proteolytic enzymes [11]. Therefore, as an environmental protection and economic pretreatment process, MP is more feasible to improve the performance of dry AD.

In addition, thermophilic conditions are often used in dry AD to obtain higher destruction rates of organic solids and superior growth of microorganisms [12]. Compared to mesophilic digestion, higher biogas production can be obtained in a shorter residence time by thermophilic digestion, which realizes a greater organic load rate. Fernández-Rodríguez, et al. [13] observed that the methane production of the organic fraction of municipal solid waste (OFMSW) at 55 ℃ was 27 % higher than that at 35 ℃, and the operation time to obtain the same methane

* Corresponding author at: College of Engineering, Northeast Agriculture University, Harbin, 150030, China.
** Corresponding author at: Guangzhou Institute of Energy Conversion, Chinese Academy of Sciences, Guangzhou, 510640, China.
E-mail addresses: kongxy@ms.giec.ac.cn (X. Kong), liwenzhe9@163.com (W. Li).

https://doi.org/10.1016/j.bej.2020.107873
Received 31 July 2020; Received in revised form 8 October 2020; Accepted 20 November 2020
Available online 25 November 2020
1369-703X/© 2020 Published by Elsevier B.V.

production was reduced from 40 to 20 days. Suhartini, et al. [14] showed that the biogas and methane productivity for thermophilic digestion of sugar beet pulp was higher than that of mesophilic digestion, and could operate at higher organic loading rates, while the latter showed signs of instability. In addition, an attention has been found that aerobic or anaerobic thermophilic bacteria have higher hydrolysis capabilities, which can be seen as a potential way to accelerate the hydrolytic process [15].

Based on the above analysis, different oxygen loads were passed into the digester for a thermophilic dry anaerobic digestion and attempted to improve the system stability and biogas production performance of thermophilic dry AD by MP. Therefore, the aims of this study were: (1) to analyze the performance of the biogas production at different microaerobic conditions; (2) to investigate the variation of VFA content and VS remove rate; (3) to explore the effects of MP on microbial community and diversity.

2. Materials and methods

2.1. Material and inoculum

Corn Stalk (CS) was collected from Zengcheng, Guangdong Province, China, which was crushed by a pulverizer (XB-HP, xiaobao,China) to obtain small particles of less than 40 meshes, and then stored in a desiccator at room temperature until further processing. The inoculum was derived from a thermophilic continuous stirred tank reactor with CS as substrates. The characteristics of raw material and inoculum are listed in Table 1.

2.2. Digester setup and operation

Anaerobic digestion experiments were performed in a 500 mL brucellae flask digester. The CS, inoculum and deionized water were placed to the digester to form 150 g of effective digestive liquid with a TS of 16.76 %. A certain proportion of the air was pumped with a vacuum pump and the same volume of pure oxygen was injected by the syringe, which created a different micro-oxygen environment. The experimental design was shown in Table 2. Finally, all digesters were operated at 55 ℃ for 35 days.

The biogas samples of AD were analyzed every day, and the analysis interval increased gradually with time. To analyze the system state of AD, samples were taken at the bottom of the digester on the 2nd, 15th and 28th days, respectively, which were used to determine the metabolites and the community succession of bacteria and archaea. All samples were stored in a refrigerator at -20 ℃ before determination and three parallels were set in each experimental group.

2.3. Analytical methods

TS and VS were determined according to the Standard method [16]. The carbon and nitrogen contents of the samples were determined by an element analyzer (Vario EL Cube, Germany). The pH value was determined by a pH meter.

The biogas content of N_2, CH_4 and CO_2 was determined by Shimadzu high-efficiency gas chromatograph (GC2014, Shimadzu, Kyoto, Japan) via a thermal conductivity detector (TCD). The VFAs contents (acetic acid, propionic acid and butyric acid) were measured by the HPLC system (waters e2695), equipped with a refractive index detector (SH-1011). The column temperature was 50 ℃ and the mobile phase was 0.005 mM H_2SO_4 with a flow rate of 0.5 mL/min.

Table 1
Compositional characteristics of CS and inoculum.

Parameters	Corn Stalk	Inoculum
TS (%)	93.26 ± 1.85	1.82 ± 0.03
VS (%)	89.06 ± 1.35	0.76 ± 0.01
pH	/	7.96 ± 0.01
C (%)	42.25 ± 0.46	/
N (%)	0.42 ± 0.01	/
C/N	101.43 ± 7.35	/

Notes: % of dry matter weight. /: not determined.

Table 2
The experimental design of AD with different microaerobic environment.

Groups	Oxygen (mL/g VS)	Inocula (mL)	Corn Stalk (g)	Deionized water (mL)
G1	0	100	25	25
G2	0.5	100	25	25
G3	1	100	25	25
G4	2	100	25	25

2.4. Analysis procedure of microbial community

16S rDNA sequencing of sample sequencing was commissioned by Sangon Biotech (Shanghai) Co., Ltd. to understand the impact of microoxygen on the ecology of digester microbial community.

2.4.1. Sampling and DNA extraction

Triplicate samples from G1 to G4 were collected on days 2, 14, and 28, and marked as G1−2d, G1−14d, G1−28d, G2−2d, G2−14d, and so on. The samples were stored at -20 ℃ for subsequent analysis. Then, the genomic DNA of the entire community was extracted with a kit (E.Z.N. A™ Mag-Bind Soil DNA Kit, OMEGA), and the extracted DNA concentration was estimated by gel electrophoresis.

2.4.2. PCR amplification and Sequencing for bioinformatics analysis

The bacterial 16S rDNA sequence fragment was amplified by universal primers (515 F and 806R, 5`-GTGCCAGCMGCCGCGGTAA-3` and 5`-GGACTACHVGGGTWTCTAAT-3`), and then the polymerase chain reaction (PCR) assays were performed on each sample. Sequencing was performed on an Illumina HiSeq 2000 (Illumina, San Diego, USA). The data was pre-processed and then separately analyzed by operational taxonomic units (OTUs) clustering and biodiversity analysis by the Ribosomal Database Project (RDP) database (http://rdp.cme.msu.edu/misc/resources.jsp).

2.5. Calculation methods

Hydrolysis is generally considered as a rate-limiting step, especially in dry AD systems with poor mass transfer. Therefore, a first-order kinetic model was used to analyze the hydrolysis rate k (day^{-1}) of the biogas production [17], as shown in Eq. (1):

$$P(t) = P_m \times \left(1 - e^{-kt}\right) \quad (1)$$

where $P_{(t)}$ represents specific biogas production for a digestion time t (mL biogas/g VS), P_m represents biogas potential of the substrate (mL biogas/g VS), k (day^{-1}) is hydrolytic rate constant and e equals to 2.718.

The modified Gompertz model has an accurate prediction analysis for methane production, maximum methane production potential and lag phase by fitting the experimental data [18], it was presented as Eq. (2):

$$G(t) = G_m \times \exp\left\{ -\exp\left[\frac{r_m \times e}{G_m}(\lambda - t) + 1\right] \right\} \quad (2)$$

where $G(t)$ represents the production of cumulative methane (mL CH_4/g VS) at time t (day), G_m is the maximum methane production potential (mL CH_4/g VS), r_m is the maximum methane production rate (mL CH_4/g VS /day), λ is the lag phase (day) while t means the time over the AD period and e equals to 2.718.

2.6. Statistical analysis

The basic data were sorted with Excel 2016 software, and the data were statistically analyzed by Origin 2017 (Origin Lab Corporation, Northampton, MA, USA).

3. Results and discussion

3.1. Biogas production performance during dry AD

Methane production is considered to be the main indicator for evaluating AD performance, and it is also the primary objective of AD to achieve higher methane production. The production and composition characteristics of biogas are shown in Fig. 1. Fig. 1(a) shows the cumulative biogas production from day AD under different microoxygen loads. Biogas production in all experimental groups was mainly concentrated 16 days before anaerobic digestion. The final values detected of G1-G4 were 199.41, 215.95, 195.54 and 176.65 mL/g VS, respectively. Correspondingly, methane content in all groups fell below 50 % after 16 days of anaerobic digestion (Fig. 1(b)). The difference was that the k of G1 is 0.1107, higher than that of all MP groups (Table 3). This suggested that MP could lead to a faster conversion of the digestion substrate to simple organic compounds due to a lower hydrolysis constant [19].

As can be seen from Fig. 1(c), the daily methane production decreased to 5 mL/g VS in all the experimental groups on the 20th day of AD, which showed that the effect of MP on the main methanogenic period of AD was very weak. During the first five days of AD, the daily methane production of G2 and G3 was significantly higher than that of G1, while that of G4 was opposite. The result was consistent with λ in Table 3, whose descending order was G4 (2.93), G1 (2.34), G2 (2.04), and G3 (1.98), respectively. This indicated that proper microaerobic made the AD have better methane-producing performance in the early stage. However, the oxygen load is important for digestive system, and excess oxygen may inhibit the activities of methanogens and reduce methane production [20]. For the cumulative methane production (Fig. 1(d)), the order observed is G2 > G3 > G1 > G4, the maximum detection was 108.31 mL/g VS, which was 13.20 % higher than that of G1 (95.68 mL/g VS). The results were similar to the study of Botheju, et al. [21], who found that methane production decreased as oxygen loads increase range 0–10.1 % (as % of feed COD).

3.2. VFA concentration and composition analysis

VFA is an important intermediate in the process of digestive substrates conversion to methane, including metabolites such as acetic acid, propionic acid and butyric acid [22]. From Fig. 2, it can be seen that the highest concentration of VFA was obtained in G3, which was 10.18 g/L and 15.87 % higher than G1 (8.79 g/L) on the second day. Whereas, the VFA concentration of G4 was 7.86 g/L, 10.58 % lower than G0. This is because higher hydrolysis is achieved in microaerobic conditions with specific ranges, which will promote VFA generation. Fu, et al. [19] also reported that excessive oxygen would consume some readily available organic substrates such as VFA. In addition, for the experimental group G1-G4, acetic acid content accounted for 67.35–72.87 % of the total VFA, indicating that the methane-producing pathway of acetic acid

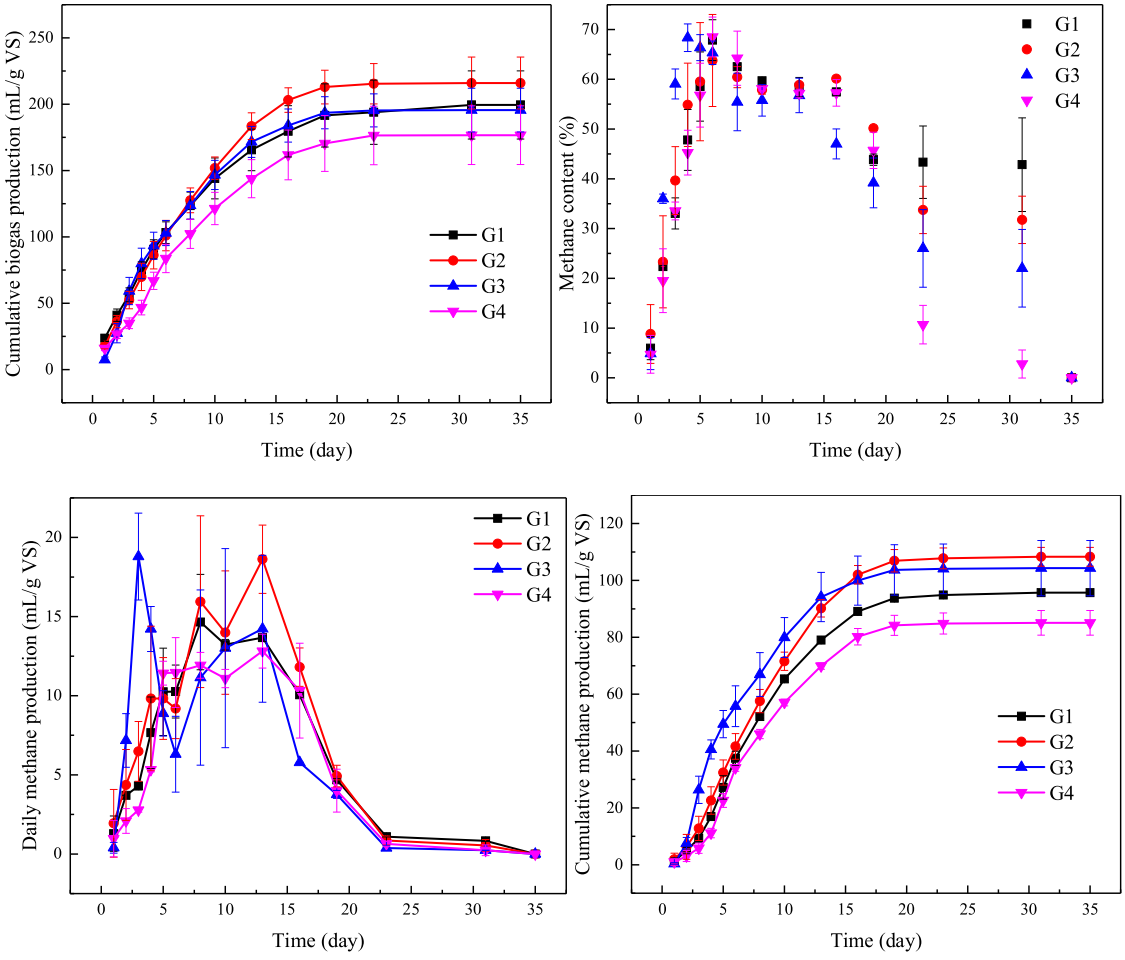


Fig. 1. Biogas production properties in AD. ((a) Cumulative biogas production, (b) Methane content, (c) Daily methane production and (d) Cumulative methane production).

Table 3
The kinetic fitting results of pretreatment with different microaerobic loads.

Groups	First-order kinetic model			Modified Gompertz model			
	P_m	k	R^2	G_m	r_m	λ	R^2
	mL Biogas /g VS	day^{-1}	–	mL CH_4 /g VS	mL CH_4 /g VS/day	day	–
G1	211.36 ± 4.61	0.11 ± 0.01	0.9975	96.40 ± 0.41	9.24 ± 9.24	2.34 ± 0.57	0.9948
G2	258.23 ± 13.26	0.08 ± 0.01	0.9913	109.36 ± 0.98	9.78 ± 0.30	1.98 ± 0.16	0.9983
G3	232.80 ± 25.49	0.08 ± 0.02	0.9596	94.97 ± 4.72	18.16 ± 1.54	2.04 ± 0.15	0.9790
G4	216.12 ± 18.54	0.07 ± 0.01	0.9783	83.11 ± 2.57	8.21 ± 0.45	2.93 ± 0.25	0.9926

Notes: -: meaningless.

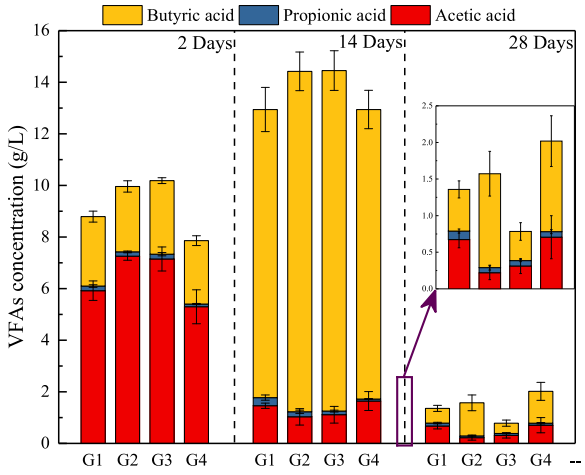


Fig. 2. Variation of VFA concentration and composition in AD.

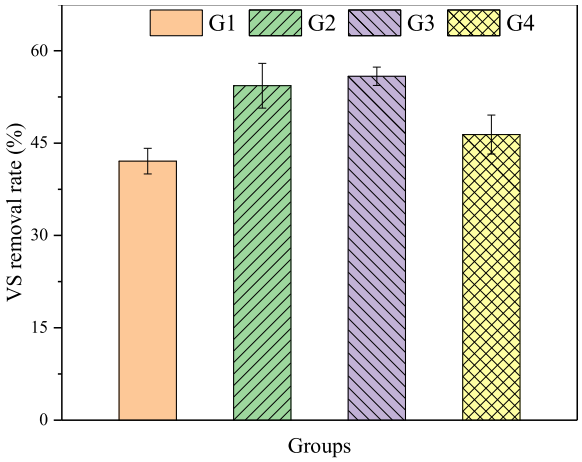


Fig. 3. The VS removal rate in AD.

played a major role in the initial stage of dry AD. On the 14th day, the VFA concentrations of G1-G4 were 12.94, 14.42, 14.45 and 12.94 g/L, respectively. It can be inferred that the digestive system may have produced inhibition due to excessive accumulation of VFA [6], and the decrease of daily methane production also supports above possibility. Among them, the contents of acetic acid were significantly reduced without accumulation. Whereas, the increased butyric acid contents exceeded 8000 mg/L, which indicated that the overaccumulation of butyric acid was the main reason for the inhibition of dry AD. On the 28th day, acid accumulation did not occur in all experimental groups with a VFA of 0.78–2.02 g/L, indicating that the substrate s was almost degraded. In addition, acetic acid and butyric acid as consumables in two methane-producing pathways are beneficial for anaerobic digestion, while the accumulation of propionic acid is a factor limiting methane production [23]. An interesting phenomenon was that the propionic acid content of G2-G4 was lower than that of G1 at day 2,14, and 28 of AD, which may be one of the reasons that MP can increase methane production.

3.3. VS removal rate

The VS removal rate is a critical indicator for evaluating AD performance, which reflects the amount of substrate recovery and environmental benefits, and its higher removal rate will result in lower environmental risks [24]. As shown in Fig. 3, the VS removal rate of G1 was 42.08 %, lower than that of the MP groups (G2-G4) by 29.08 %, 32.75 % and 10.22 %, respectively. It should be noted that the substrates and inocula of AD were identical prior to the initial and final VS measurements in this study. Therefore, other factors on the analysis of VS removal rate can be ignored.

3.4. Analysis of microbial diversity and community

AD is a complex multi-stage biochemical reaction. As the executor of the digestion process, the diversity and abundance of microorganisms indirectly reflect the running state of digestive system [25]. During digestion, microbial diversity and community vary with the change of digestion parameters and time [26]. Therefore, the diversity and structure of bacteria and archaea under different microaerobic conditions on days 2,15, and 28 were analyzed in this study.

As showed in Table 4, the sequence reads for G1-G4 were 39,535−75,610 for bacteria and 49,700−83,984 for archaea, with a total of 1,024−3,025 and 511–742 OTUs identified for bacteria and archaea, respectively. In addition, some related diversity indexes are also listed in this table. The Chao1 index for OTUs evaluation was 1,733.54−3,558.08 for bacteria and 1,437.20–3285.37 for archaea. The indexes (Shannon and Simpson) for microbial diversity evaluation were 1.98–4.41 and 0.03−0.27 for bacteria, respectively, and 2.08–3.01 and 0.13−0.22 for archaea, respectively. All coverage values greater than 0.99 indicated that most of the microbial community in the reactors had been detected.

3.4.1. Effects of microaerobic pretreatment on bacterial community

The community of bacteria at phylum level are shown in Fig. 4(a). The phyla with relative abundances greater than 1% in at least one sample of all the experimental groups were *Firmicutes*, *Bacteroidetes*, *Synergistetes* and *Proteobacteria*. *Firmicutes* dominated the entire AD process for all experimental groups, which produces extracellular enzymes (i.e. cellulase, lipase and protease) that promotes the degradation of cellulose, fat and proteins [27]. *Bacteroidetes* play an important role in the degradation and VFA generation of organic compounds such as carbohydrates and proteins [28]. Some acetogens in *Synergistic* are involved in the conversion of VFA to acetate during AD process, and hydrogen can be generated to feed the hydrogenotrophic methanogens [29]. Proteobacteria are also capable of hydrolysis and acidogenesis in AD [24].

On the second day, there was no significant difference in phylum level for all the experimental groups due to the initial AD, and *Firmicutes*, as the main phylum, had a relative abundance of 96.35 %, 92.18 %,

Table 4
The diversity analysis of bacteria and archaea.

sample		Sequence Number	OUTs number	Chao1 index	Shannon index	Simpson index	Coverage
G1−02d	Bacteria	42745	1024	1733.54	3.48	0.10	0.99
G1−15d		69437	1650	2723.36	3.91	0.19	0.99
G1−28d		71581	1805	2896.34	2.74	0.23	0.99
G2−02d		46818	1283	2263.29	4.25	0.04	0.99
G2−15d		60454	2669	3214.63	4.34	0.10	0.99
G2−28d		68902	3025	3558.08	2.96	0.19	0.99
G3−02d		46263	1245	1944.46	3.69	0.08	0.99
G3−15d		46483	1066	2283.26	3.71	0.12	0.99
G3−28d		72516	2028	2891.94	2.17	0.19	0.99
G4−02d		39535	1308	2244.62	4.41	0.03	0.99
G4−15d		55370	1033	2075.44	3.15	0.20	0.99
G4−28d		75610	1982	2452.65	1.98	0.27	0.99
G1−02d	Archaea	49700	511	1466.92	2.08	0.22	0.99
G1−15d		72247	629	2724.10	2.25	0.16	1.00
G1−28d		54225	603	2961.02	3.01	0.14	1.00
G2−02d		54068	531	1437.20	2.27	0.19	0.99
G2−15d		65585	742	3051.55	2.44	0.16	1.00
G2−28d		59694	689	3200.96	3.25	0.13	1.00
G3−02d		60305	572	1547.21	2.25	0.19	0.99
G3−15d		69016	728	3130.15	2.33	0.16	1.00
G3−28d		58624	702	3285.37	2.91	0.14	0.99
G4−02d		58142	506	1902.07	2.54	0.13	0.99
G4−15d		83984	667	2571.05	2.24	0.17	1.00
G4−28d		59021	688	2931.02	2.84	0.14	0.99

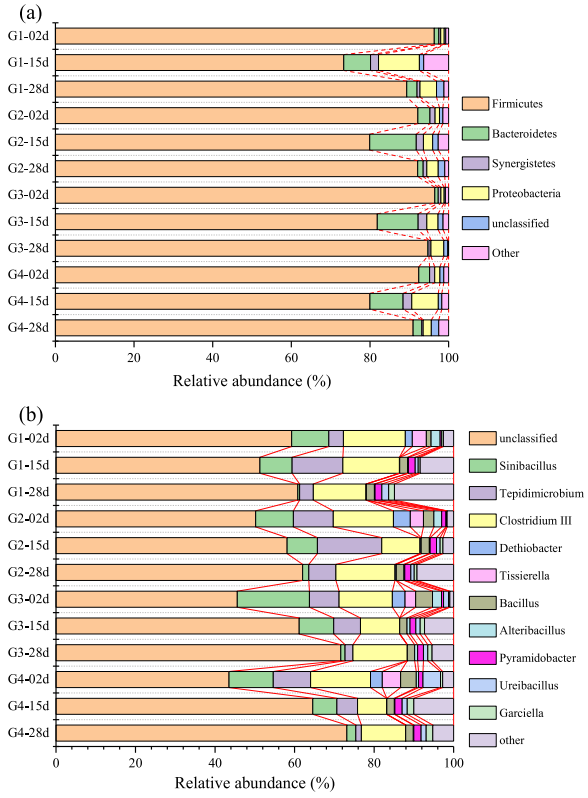


Fig. 4. The relative abundance of bacterial communities on phylum level (a) and genus level (b).

96.49 % and 92.42 %, respectively. Subsequently, for all the experimental groups, the relative abundance of *Firmicutes* decreased, while that of *Bacteroidetes* increased significantly on the 15th day. It can be inferred that more acidogenic bacteria are involved in the metabolic activities of AD than the second day. In addition, the relative abundance of *Firmicutes* in G2-G4 exceeded that of G0 (73.32 %) by 9.02%–11.65%. This was because more oxygen was provided to support the growth and metabolism of facultative bacteria under microaerobic conditions [30]. It also promoted the production of more hydrolases, which meant that the MP has a higher hydrolysis rate. On the 15th day, the relative abundance of *Firmicutes* increased again while that of *Bacteroidetes* decreased, indicating that the acidogens activity was weakened in AD process, which was consistent with the changing trend of VFA concentration.

The relative abundance changes of bacterial communities at generic level in each experimental group are shown in Fig. 4(b). ten genera with a relative abundance greater than 1% were detected, dominated by *Sinibacillus*, *Tepidimicrobium* and *Clostridium*. *Sinibacillus* is a typical thermotropic facultative anaerobic bacteria, which is the dominant bacteria that degrades organic matter in thermotropic AD [31]. For the all experimental groups, the relative abundance of *Sinibacillus* gradually decreased on days 2, 15, and 28. It is not difficult to understand that the substrate hydrolysis process is mainly concentrated in the early AD. Differently, on the second day of AD, the relative abundance of G1 was only 9.8 %, lower than MP groups, whose relative abundance was 9.85 % (G2), 19.04 % (G3) and 11.68 % (G4), respectively. The results also confirmed that the AD with MP had a better hydrolysis rate. The abilities of *Tepidimicrobium* are responsible for degrading cellulose and producing VFA [32]. Generally, the amount of *Tepidimicrobium* is considered as an indicator for the acidification process [33]. *Clostridium* is also a genus with the strong capacity of hydrolysis and acid production [34]. Both of them are responsible for the conversion of molecular organic matter into volatile fatty acids. Their relative abundance in G2-G4 was 26.19 %, 21.91 % and 25.68 %, respectively, higher than that of G1 on the second day, while the opposite was the case on the 15th day. The relative abundance of G1 was 28.34 %, higher than that of the MP groups from 10.01–115.35%. The results show that MP has a higher acidogenesis rate in the early AD phase.

3.4.2. Effects of microaerobic pretreatment on archaeal community

Methanogenic archaea are generally considered to be strictly anaerobic, and oxygen intervention may be disastrous for the activity of methanogenic methanogens [35]. Therefore, it is necessary to explore the effects of MP on the community of archaea in AD. As can be seen from Fig. 5, all classified archaea from the four experimental groups were affiliated with six major methanogenic genera, including one acetoclastic (*Methanothrix*) and five hydrogenotrophic (*Methanobacterium, Methanomassiliicoccus, Methanosphaera, Methanobrevibacter* and *Methanolinea*,) methanogens [36,37]. On the whole, the

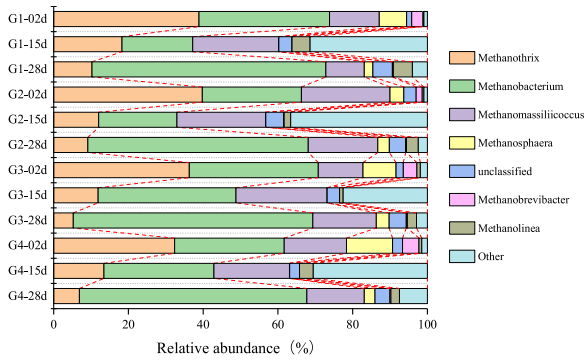


Fig. 5. The relative abundance of archaeal community on genus level.

relative abundance of aceticlastic methanogens in all the experimental groups decreased gradually, while that of hydrogenotrophic methanogens increased gradually with the anaerobic digestion time, and the change trend of G1 was more obvious. This indicated that the hydrogenotrophic methanogenic pathway was more advantageous in dry AD, and MP further promoted this transformation [38]. Chen, et al. [39] also mentioned that MP could remove sulfide and accelerate the growth of hydrogenotrophic methanogens to improve AD performance. In addition, genus *Methanobacterium*, as a dominant hydrogenotrophic methanogen in this study, has been reported to have limited oxygen tolerance [40]. This may be the reason why the relative abundance of *Methanobacterium* in the MP group was higher than that of G1 on the 15th days.

4. Conclusion

The present study confirmed that MP could improve the system stability and biogas production performance of the thermophilic dry AD. The results showed that proper oxygen injection promoted the generation of VFA during hydrolysis acidogenesis stage and reduced propionic acid content throughout the digestive cycle. After digestion, the maximum cumulative methane production of 108.31 mL/g VS was obtained at G3, higher than G1 (95.68 mL/g VS) 13.2 %, and higher VS removal rate was obtained at the same oxygen load (5 mL/g VS). In addition, microbial community diversity indicated that the relative abundance of *Firmicutes* was increased by MP, which facilitated the hydrolysis of CS substrate, and the hydrogenotrophic methanogenic pathway was strengthened to improve AD performance.

Authors contribution

Prof. Feng Zhen: He has conceptualization, design and supervision of review and also writing - original draft this article. In addition, funding acquisition and project administration.

Dr. Xinjian Luo: He is supervision, conceptualization and data correction.

Prof. Tao Xing: He has put some contribution to data correction and formal analysis, as well as very help to improve the manuscript quality and English.

Prof. Yongming Sun: Prof. Yongming Sun is supervision, validation, and visualization, as well as very help to improve the manuscript quality and English.

Prof. Xiaoying Kong: He has conceptualization, design and supervision of review and also writing - original draft this article.

Prof. Wenzhe Li: He has conceptualization, design and supervision of review and also writing - original draft this article. In addition, funding acquisition and project administration.

Declaration of Competing Interest

The authors report no declarations of interest.

Acknowledgements

The authors are grateful for the support of National Natural Science Foundation of China (21978289), Chinese Academy of Sciences Strategic Leading Science and Technology Project (Class A) (XDA21050400), Guangdong Province Science and Technology Project (2017B020238005), Key R&D Project of Jiangxi Province (20171ACG70020), Major research and development projects in Jiangxi Province (20182ABC28006) and Guangzhou Science and Technology Project (201803030007).

References

[1] J. Jiang, L. Li, Y. Li, Y. He, C. Wang, Y. Sun, Bioaugmentation to enhance anaerobic digestion of food waste: dosage, frequency and economic analysis, Bioresour. Technol. 307 (2020), 123256.
[2] F. Zhen, J. Zhang, W. Li, Y. Sun, X. Kong, Optimizing waste heat utilization in vehicle bio-methane plants, Energies 11 (2018).
[3] Y. Sun, Z. Zhang, Y. Sun, G. Yang, One-pot pyrolysis route to Fe−N-Doped carbon nanosheets with outstanding electrochemical performance as cathode materials for microbial fuel cell, Int. J. Agric. Biol. Eng. (2020), https://doi.org/10.25165/j.ijabe.20201306.5765.
[4] Y. Li, R. Zhang, C. Chen, G. Liu, Y. He, X. Liu, Biogas production from co-digestion of corn stover and chicken manure under anaerobic wet, hemi-solid, and solid state conditions, Bioresour. Technol. 149 (2013) 406–412.
[5] X. Ge, F. Xu, Y. Li, Solid-state anaerobic digestion of lignocellulosic biomass: recent progress and perspectives, Bioresour. Technol. 205 (2016) 239–249.
[6] I. Rocamora, S.T. Wagland, R. Villa, E.W. Simpson, O. Fernández, Y. Bajón-Fernández, Dry anaerobic digestion of organic waste: A review of operational parameters and their impact on process performance, Bioresour. Technol. 299 (2020), 122681.
[7] I. Ramos, M. Fdz-Polanco, The potential of oxygen to improve the stability of anaerobic reactors during unbalanced conditions: results from a pilot-scale digester treating sewage sludge, Bioresour. Technol. 140 (2013) 80–85.
[8] P. Tsapekos, P.G. Kougias, S.A. Vasileiou, G. Lyberatos, I. Angelidaki, Effect of micro-aeration and inoculum type on the biodegradation of lignocellulosic substrate, Bioresour. Technol. 225 (2017) 246–253.
[9] J.W. Lim, J.A. Chiam, J.-Y. Wang, Microbial community structure reveals how microaeration improves fermentation during anaerobic co-digestion of brown water and food waste, Bioresour. Technol. 171 (2014) 132–138.
[10] J.W. Lim, J.-Y. Wang, Enhanced hydrolysis and methane yield by applying microaeration pretreatment to the anaerobic co-digestion of brown water and food waste, Waste Manag. 33 (2013) 813–819.
[11] S.-F. Fu, S. He, X.-S. Shi, N.R. Katukuri, M. Dai, R.-B. Guo, The chemical properties and microbial community characterization of the thermophilic microaerobic pretreatment process, Bioresour. Technol. 198 (2015) 497–502.
[12] J.K. Kim, B.R. Oh, Y.N. Chun, S.W. Kim, Effects of temperature and hydraulic retention time on anaerobic digestion of food waste, J. Biosci. Bioeng. 102 (2006) 328–332.
[13] J. Fernández-Rodríguez, M. Pérez, L.I. Romero, Comparison of mesophilic and thermophilic dry anaerobic digestion of OFMSW: kinetic analysis, Chem. Eng. J. 232 (2013) 59–64.
[14] S. Suhartini, S. Heaven, C.J. Banks, Comparison of mesophilic and thermophilic anaerobic digestion of sugar beet pulp: performance, dewaterability and foam control, Bioresour. Technol. 152 (2014) 202–211.
[15] C. Dumas, S. Perez, E. Paul, X. Lefebvre, Combined thermophilic aerobic process and conventional anaerobic digestion: effect on sludge biodegradation and methane production, Bioresour. Technol. 101 (2010) 2629–2636.
[16] Z. Wang, Z. Liu, R.S. Noor, Q. Cheng, X. Chu, B. Qu, F. Zhen, Y. Sun, Furfural wastewater pretreatment of corn stalk for whole slurry anaerobic co-digestion to improve methane production, Sci. Total Environ. 674 (2019) 49–57.
[17] D. Li, X. Huang, Q. Wang, Y. Yuan, Z. Yan, Z. Li, Y. Huang, X. Liu, Kinetics of methane production and hydrolysis in anaerobic digestion of corn stover, Energy 102 (2016) 1–9.
[18] B. Shamurad, N. Gray, E. Petropoulos, S. Tabraiz, E. Membere, P. Sallis, Predicting the effects of integrating mineral wastes in anaerobic digestion of OFMSW using first-order and Gompertz models from biomethane potential assays, Renew. Energy 152 (2020) 308–319.
[19] S.-F. Fu, F. Wang, X.-Z. Yuan, Z.-M. Yang, S.-J. Luo, C.-S. Wang, R.-B. Guo, The thermophilic (55°C) microaerobic pretreatment of corn straw for anaerobic digestion, Bioresour. Technol. 175 (2015) 203–208.
[20] S. Xu, A. Selvam, J.W.C. Wong, Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste, Waste Manag. 34 (2014) 363–369.
[21] D. Botheju, G. Samarakoon, C. Chen, R. Bakke, An experimental study on the effects of oxygen in bio-gasification-part 2, International Conference on Renewable Energies and Power Quality (ICREPQ'10) (2010).

[22] Z. Wang, Q. Cheng, Z. Liu, J. Qu, X. Chu, N. Li, R.S. Noor, C. Liu, B. Qu, Y. Sun, Evaluation of methane production and energy conversion from corn stalk using furfural wastewater pretreatment for whole slurry anaerobic co-digestion, Bioresour. Technol. 293 (2019), 121962.
[23] Y. Chen, H. Yang, Z. Zhao, H. Zou, R. Zhu, Q. Jiang, T. Sun, M. Li, L. Li, D. Shi, H. Ai, Q. He, L. Gu, Comprehensively evaluating the digestive performance of sludge with different lignocellulosic components in mesophilic anaerobic digester, Bioresour. Technol. 293 (2019), 122042.
[24] Y. Chen, R. Zhu, Q. Jiang, T. Sun, M. Li, J. Shi, H. Chai, L. Gu, H. Ai, Q. He, Effects of green waste participation on the co-digestion of residual sludge and kitchen waste: a preliminary study, Sci. Total Environ. 671 (2019) 838–849.
[25] A. Sepehri, M.-H. Sarrafzadeh, Effect of nitrifiers community on fouling mitigation and nitrification efficiency in a membrane bioreactor, Chem. Eng. Process. Process. Intensif. 128 (2018) 10–18.
[26] D. Li, C. Jiao, W. He, Z. Yan, Y. Yuan, Z. Li, Y. Guo, X. Liu, Comparison of micro-aerobic and anaerobic fermentative hydrogen production from corn straw, Int. J. Hydrogen Energy 41 (2016) 5456–5464.
[27] M. Wang, X. Zhang, J. Zhou, Y. Yuan, Y. Dai, D. Li, Z. Li, X. Liu, Z. Yan, The dynamic changes and interactional networks of prokaryotic community between co-digestion and mono-digestions of corn stalk and pig manure, Bioresour. Technol. 225 (2017) 23–33.
[28] X. Guo, C. Wang, F. Sun, W. Zhu, W. Wu, A comparison of microbial characteristics between the thermophilic and mesophilic anaerobic digesters exposed to elevated food waste loadings, Bioresour. Technol. 152 (2014) 420–428.
[29] D.H. Lee, S.K. Behera, J.W. Kim, H.-S. Park, Methane production potential of leachate generated from Korean food waste recycling facilities: a lab-scale study, Waste Manag. 29 (2009) 876–882.
[30] W. Charles, L. Walker, R. Cord-Ruwisch, Effect of pre-aeration and inoculum on the start-up of batch thermophilic anaerobic digestion of municipal solid waste, Bioresour. Technol. 100 (2009) 2329–2335.
[31] Y. Yang, K. Yu, Y. Xia, F.T.K. Lau, D.T.W. Tang, W.C. Fung, H.H.P. Fang, T. Zhang, Metagenomic analysis of sludge from full-scale anaerobic digesters operated in municipal wastewater treatment plants, Appl. Microbiol. Biotechnol. 98 (2014) 5709–5718.
[32] E. Kim, J. Lee, G. Han, S. Hwang, Comprehensive analysis of microbial communities in full-scale mesophilic and thermophilic anaerobic digesters treating food waste-recycling wastewater, Bioresour. Technol. 259 (2018) 442–450.
[33] L. Regueiro, J.M. Lema, M. Carballa, Key microbial communities steering the functioning of anaerobic digesters during hydraulic and organic overloading shocks, Bioresour. Technol. 197 (2015) 208–216.
[34] A.M. Ziganshin, J. Liebetrau, J. Pröter, S. Kleinsteuber, Microbial community structure and dynamics during anaerobic digestion of various agricultural waste materials, Appl. Microbiol. Biotechnol. 97 (2013) 5161–5174.
[35] S.-F. Fu, F. Wang, X.-S. Shi, R.-B. Guo, Impacts of microaeration on the anaerobic digestion of corn straw and the microbial community structure, Chem. Eng. J. 287 (2016) 523–528.
[36] Y. Ren, H. Hao Ngo, W. Guo, D. Wang, L. Peng, B.-J. Ni, W. Wei, Y. Liu, New perspectives on microbial communities and biological nitrogen removal processes in wastewater treatment systems, Bioresour. Technol. 297 (2020), 122491.
[37] G. Capson-Tojo, R. Moscoviz, S. Astals, Á. Robles, J.P. Steyer, Unraveling the literature chaos around free ammonia inhibition in anaerobic digestion, Renewable Sustainable Energy Rev. 117 (2020), 109487.
[38] M. Zhou, H. Yang, D. Zheng, X. Pu, Y. Liu, L. Wang, Y. Zhang, L. Deng, Methanogenic activity and microbial communities characteristics in dry and wet anaerobic digestion sludges from swine manure, Biochem. Eng. J. 152 (2019), 107390.
[39] Q. Chen, W. Wu, D. Qi, Y. Ding, Z. Zhao, Review on microaeration-based anaerobic digestion: state of the art, challenges, and prospectives, Sci. Total Environ. 710 (2020), 136388.
[40] B. Deshai, B. Rune, Oxygen effects in anaerobic digestion–A review, Open Waste Manag. J. 4 (2011) 1–19.